NULL TYPE RECORDERS

A Null type recorder is a type of strip chart recorder. These recorders are based on the principle of null condition or self balancing principle.

This type of recorder consists of a measuring circuit which is at balanced condition. The circuit becomes imbalanced when an input signal is applied to it. Due to the imbalance created by the input signal an error signal is generated. The error signal is the difference between the input signal and a reference signal. Hence, the measuring circuit used in a null type recorder is a comparison circuit. The error signal is used to operate an instrument which balances the circuit. The balancing instrument is usually a type of moving mechanism which moves the wiper in such a direction which reduces the error and balances the circuit. The balancing instrument moves by an amount proportional to the magnitude of the error. The recording pen is connected to the balancing instrument in such a way that the signal gets recorded during the balance of the circuit i.e the recording pen moves in synchronism with the wiper. The recording ~~process~~ comes to an end when the null condition is attained (i.e when error = 0). The most commonly used null type recorders are of three types. They are:

1. Potentiometric recorder
2. Bridge type
3. ~~B~~LVDT recorder (Linear Variable Differential Transformer)

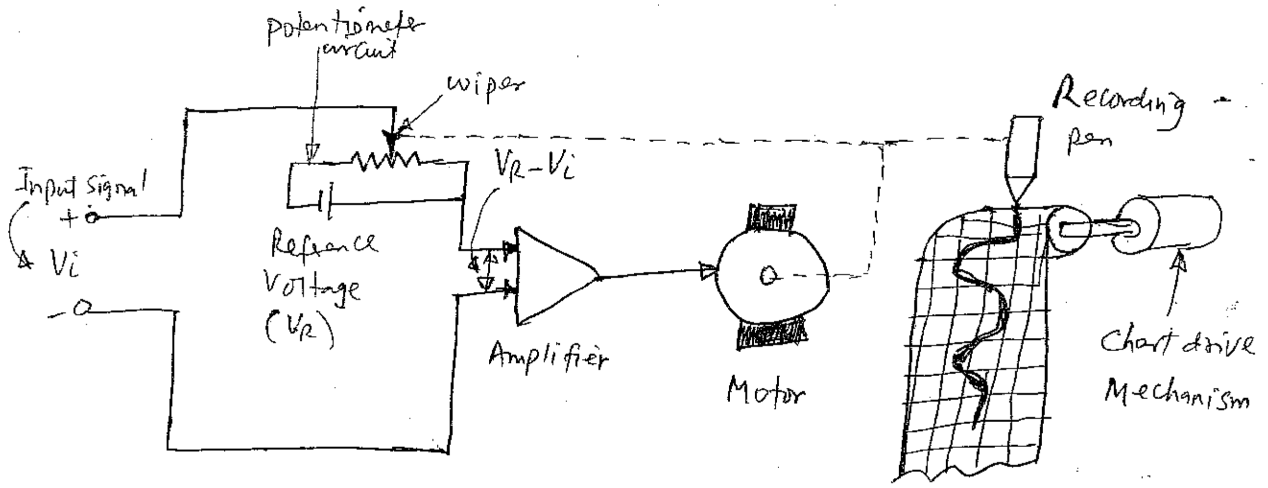

Figure potentiometric Recorder

In potentiometric recorder, the principle of recording (i.e null type condition or self balancing) is implemented by comparing the input voltage to be recorded with the refrence voltage of a potentiometer. The comparison (diffrence) between the input voltage and refrence voltage results in an error voltage. The error voltage is then amplified by an op-amp and applied as an input to the D.C motor.

The wiper of the potentiometer is mechanically connected the armature of the D.C motor therefore the rotation of the armature causes the movement of wiper in the direction which decreases the amount of error signal. When the error becomes zero there will be no supply to the D.C motor due to which the armature of the motor stops and consequently the wiper also comes to rest and thus a null condition is achieved.

As the recording pen is connected to the wiper, the recording pen moves over the rolling graph paper by an amount and direction same as that of the wiper. In potentiometric recorders, the chart (graph paper) is moved by chart drive mechanism which is synchronized to the power line frequency. Thus, in the process of balancing the input signal against the potentiometer voltage, the input signal gets recorded.

Bridge Type recorder

CONSTRUCTION

A block diagram of a bridge type recorder is shown in figure below.

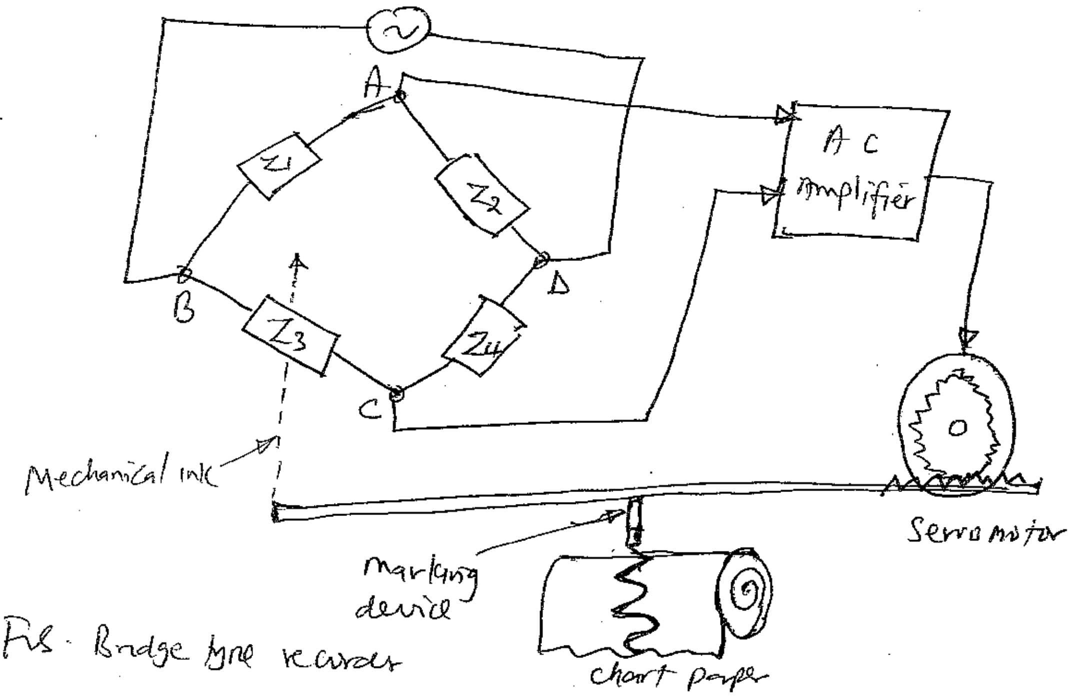

Fig. Bridge type recorder

In this recorder a servomotor is employed for balancing the bridge. One of the arms of the ac bridge has a variable impedance, resistance, inductance or capacitance as per requirement of the bridge. This impedance varies as the servomotor rotates because both of these are linked mechanically. Other arm of the ac bridge is a transducer in which some electrical parameter i.e. R, L or C varies with the variation in input signal under monitoring. With the variation in input signal the value of impedance Z_1 changes. So the ac bridge gets unbalanced and ac voltage appears across points A and C, which is supplied to the servomotor through an ac amplifier. Servomotor starts rotating and produces a change in the value of impedance Z_3 such that ac bridge gets balanced again. In the balanced condition of the bridge no voltage appears across the terminals ~~A~~ A and C and so the servomotor stops rotating.

Degree of rotation of servo-motor depends on how much change has occurred in the value of impedance Z_1 as with the change in the value of Z_1, the servomotor varies the value of Z_3 to keep the ratio of Z_1 and Z_3 constant for making the bridge balanced.

Bridge recorders are used when the signals to be monitored are due to variation in some electrical ~~parameters~~ of a passive transducer such as variation ~~of~~ in resistance which occur in strain gauges, thermistors, and photoconductive cells.

LINEAR VARIABLE DIFFERENTIAL TRANSFORMER (LVDT) RECORDERS

This type of recorder is used when signals to be monitored are manifested in the form of small displacements. In this recorder LVDTs are used to generate a self-balancing capability as shown in figure below.

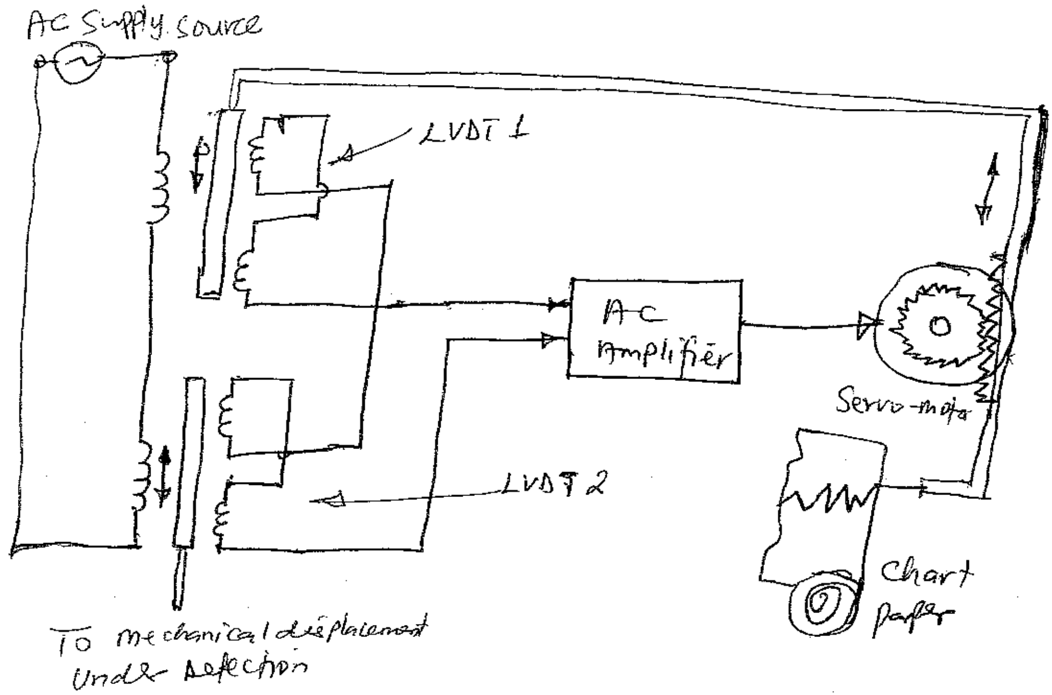

figure LVDT type recorder

Moving core of LVDT 2 is linked with the sensor of which mechanical movement to be recorded. Moving core of LVDT 1 is linked mechanically with shaft of the servomotor. Output of both of the LVDTs are connected in series opposition. When the core of the LVDT 2 moves, output voltage of LVDT 2 differs from the output voltage of LVDT 1 and the difference of these two voltages appears at the input of ac amplifier. AC amplifier amplifies the input voltage and supplies it to the servomotor. Then the servomotor rotates in such a direction that it moves the core of LVDT 1 so that output voltage of LVDT 1 becomes equal to output voltage of LVDT 2. Now the input voltage to the ac amplifier becomes again zero and the servo-motor stops rotating.

2. GALVANOMETER TYPE RECORDERS (2)

The D'Arsonval movement used in moving coil indicating instruments Can also Can also provide the movement in a galvanometer type recorder.

The D'Arsonval movement consists of a moving coil placed in a strong Magnetic field, as shown in figure below.

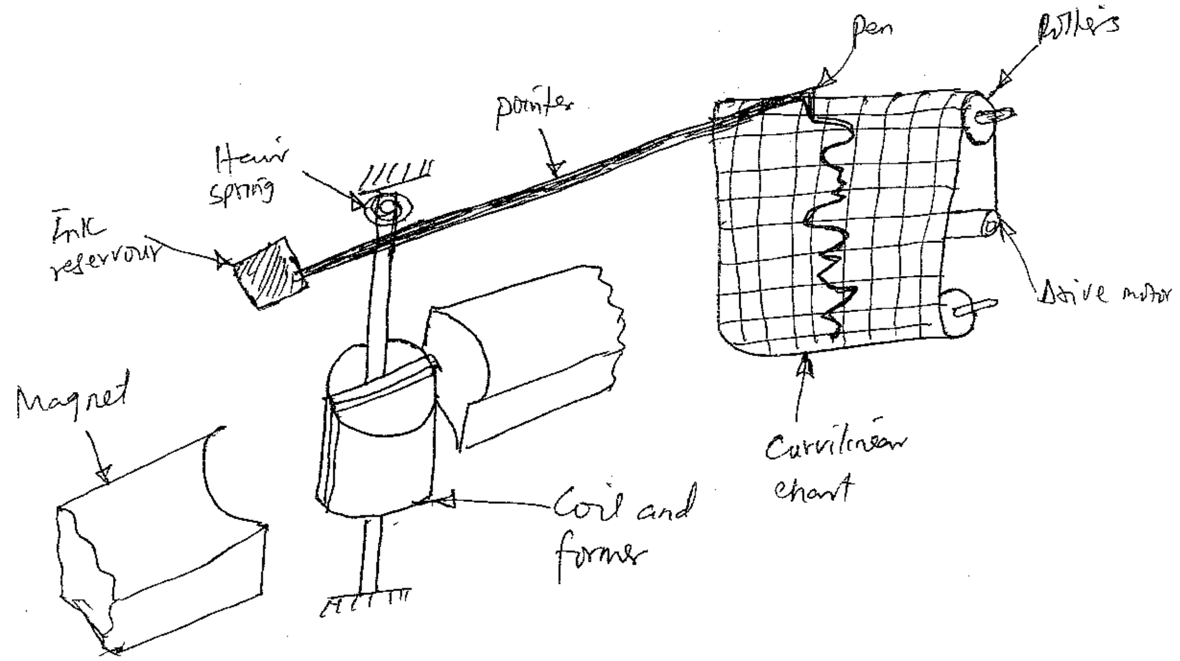

Figure Galvanometer type recorder

In a galvanometer type recorder, the pointer of the D'Arsonval movement is fitted with a pen-ink (Stylus) mechanism. The pointer deflects when current flows through the moving coil. The deflection of the pointer is directly proportional to the magnitude of the current flowing through the coil.

As the signal current flows through the coil, the magnetic field of the coil varies in intensity in accordance with the signal. The reaction of this field with the field of the permanent Magnet causes the coil to change its angular position. As the position of the coil follows the variation of the signal current being recorded, the pen is accordingly deflected across the paper chart.

The paper is pulled from a supply roll by a motor driven transport mechanism. Thus as the paper moves past the pen and as the pen is deflected, the signal waveform is traced on the paper.

The recording pen is connected to an ink reservour through a narrow bore tube. Gravity and capillary action establishes a flow of ink from the reservour through the tubing and into the hollow of the pen.

OSCILLOGRAPH RECORDERS

These recorders are instruments which record information on paper or film with ink, powder, electrostatic change, Liquid, electric arc or light. Oscillograph implies either a galvanometer or a CRT recorder.

1. GALVANOMETER OSCILLOGRAPHIC RECORDERS

Such recorders are two dimensional display and recording devices having a mirror galvanometer, Light source and light sensitive recording surface which rolls up continously. The basic operation of such recorder is as shown in figure below.

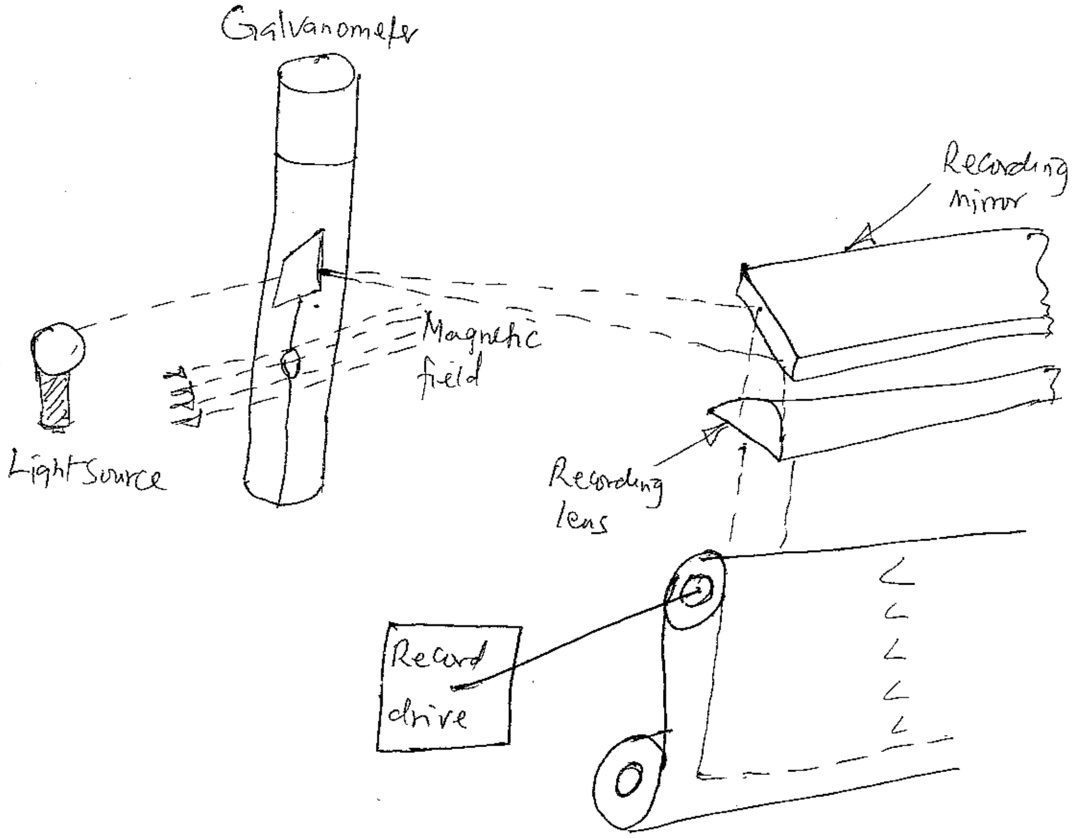

Figure Galvanometer Oscillographic Recorder

Light is reflected from the galvanometer mirror and falls on to a recording Surface. The light beam moves in a Straight line on the recording Surface in proportion to the amplitude and the rate of change in the output signal being recorded.

So this way amplitude Vs time record of input signal is provided. High pressure Mercury Vapour lamp is usually used an ultraviolet light Source.

Refrence grid lines are recorded by passing a portion of light through a grid line aperture which allows only series of fine bars of light running in the same direction as the record travel, and exposed on the record simultaneously with the galvanometer traces.

Refrence timing lines are recorded by a timing circuit which activates a flasher momentarily and these lines are recorded on the paper across the record and perpendicular to the grid.

Refrence grid lines are recorded by passing a portion of light through a grid line aperture which allows only series of fine bars of light running in the same direction as the record travel, and exposed on the record simultaneously with the galvanometer traces.

Refrence timing lines are recorded by a timing circuit which activates a flashes momentarily and these lines are recorded on the paper across the record and perpendicular to the grid.

CRT RECORDERS

These are four dimensional display and recording devices. Two axes out of the four axes are conventional ones i.e X-axis and Y-axis. Its third axis is Z-axis which is defined by spot intensity on display media or recording surface. Its fourth axis i.e Y'-axis is movement of recording medium. The block diagram of a CRT recorder is shown in figure below.

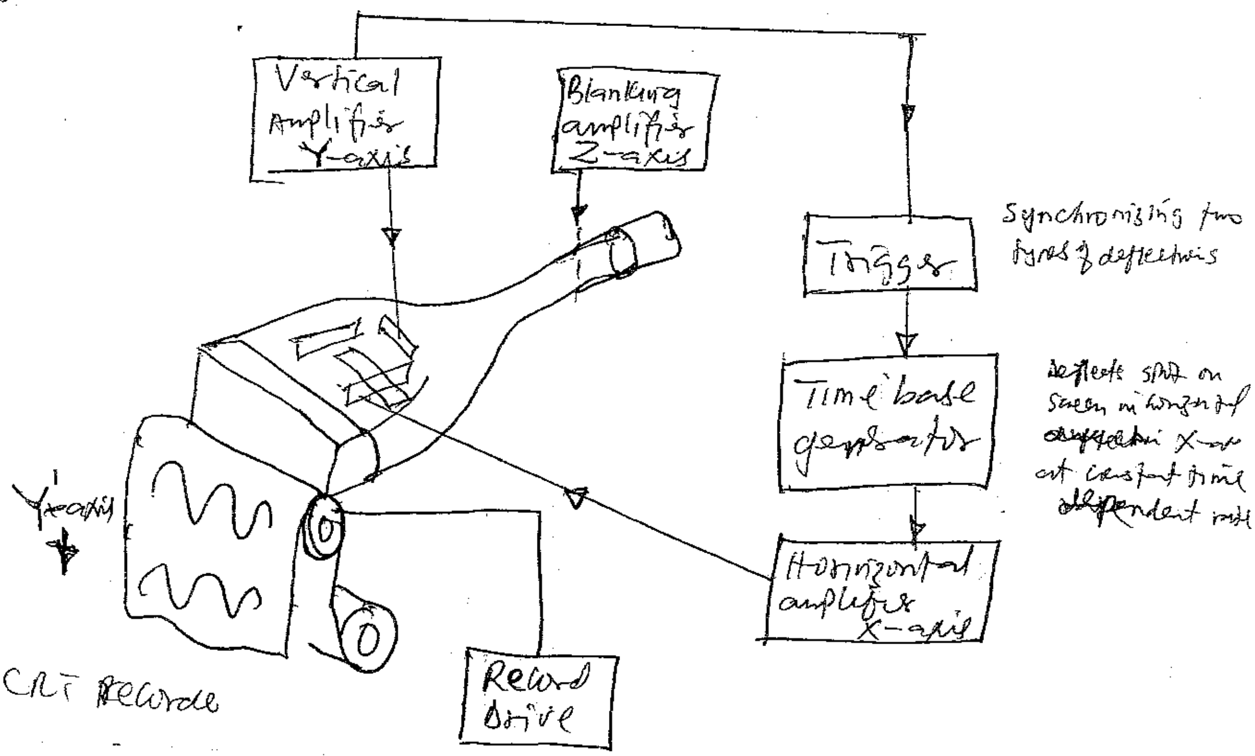

Figure CRT Recorder

OSCILLOGRAPH RECORDERS

These recorders are instruments which record information on paper or film with ink, ~~[illegible]~~ powder, electrostatic charge, liquid, electric arc or light. Oscillograph implies either a galvanometer or a CRT recorder - the recorders having bandwidths more than 20KHz.

1. GALVANOMETER RECORDERS

Such recorders are two dimensional display and recording devices having a mirror galvanometer, light source and a light sensitive recording surface, which rolls up continuously. The basic operation of such a recorder is shown below.

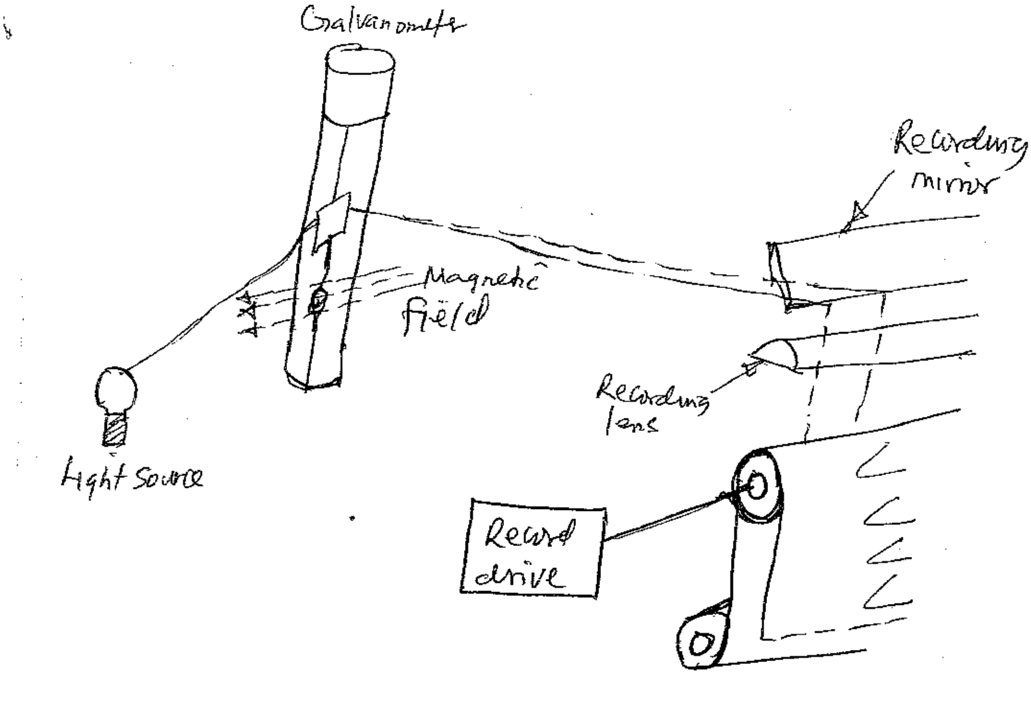

Figure Galvanometer recorder

Light is reflected from the galvanometer mirror and falls on to a recording device surface. So the light beam moves in a straight line on the recording surface, in proportion to the amplitude and to the rate of change in the output signal being recorded. So this way amplitude vs time record of input signal is provided. High pressure mercury vapour lamp is usually used as an ultra-violet light source.

CRT

As the name implies a cathode ray tube is used and its output light is focused on the record either by the optical lens system or by providing a fibre optic face plate on the CRT and placing the record in contact with the fibre optics.

Reference grid lines in the horizontal and vertical directions are recorded by putting the light from the flash tube through an aperture. The timing of the flash tube can be controlled and synchronised with the record drive system.

MAGNETIC TAPE RECORDERS

A magnetic tape recorder consists of the following basic elements:

(a) The record head that responds to an electrical signal and creates a magnetic pattern on a magnetizable medium.

(b) The magnetic tape, as the magnetizable medium, that conforms to and retains the magnetic pattern.

(c) The reproduce head that detects a magnetic pattern on the tape and converts it into an electrical signal.

(d) Conditioning devices, such as amplifiers and filters, to modify the input signal to a format that can be properly recorded on the tape.

(e) A tape transport mechanism that moves the tape along the record and reproduce heads at a constant speed.

A simplified block diagram of a magnetic recording system is shown in Figure below.

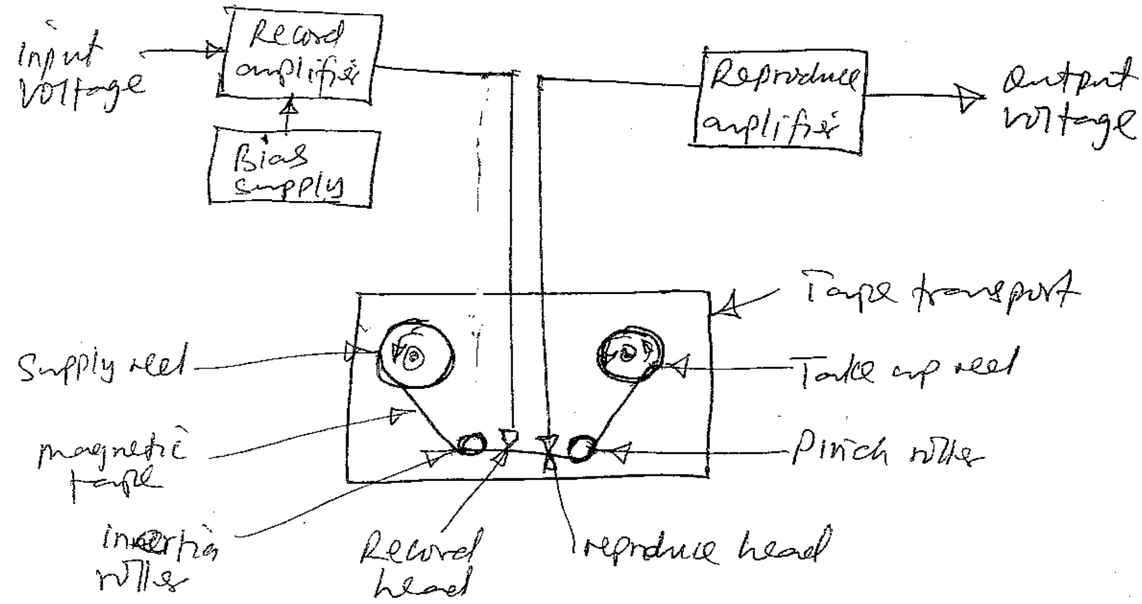

Figure Simple Magnetic recording system.

Three recording methods are specified to meet the various industrial requirements as follows:
(i) Direct recording (ii) Frequency modulation or FM recording
(iii) pulse modulation or PM recording.

DIRECT RECORDING

A recording head is similar to a toroidal transformer with a single winding is shown in Figure